Kevin Babakhani

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EDUCATION

California State Polytechnic University, Pomona - B.S, Computer Science December 2024 - 3.83 GPA

Coursework: Software Engineering, Data Structures and Algorithms, Object-Oriented Programming, Computer Networks, Mobile Application Development, Game Development, Artificial intelligence, Operating Systems

Awards and Involvement: Summa Cum Laude, Presidents List (2024), CPP Dean's List (2023-2024)

WORK EXPERIENCE

Coding Instructor - Coding Mind Academy - Remote October 2024 - Present

- Teach 10+ K–12 students programming concepts and projects using **Python, Java, JavaScript, HTML/CSS, and AI/prompt-engineering tools.**
- Design interactive lessons for different age groups and skill levels; lead hands-on projects that develop students' coding and problem-solving skills.
- Assess student progress and provide individualized feedback; student evaluations indicate a 25% improvement in comprehension and retention.

Operations Attendant - Associated Students Incorporated (CPP) - Pomona CA January 2023 - December 2024

- Managed an inventory system for 500+ student equipment checkouts, improving organization and automating processes.
- Collaborated on task assignments and workflow improvements to reduce operational bottlenecks and improve response times.
- Implemented safety protocols and responded to 3+ emergency incidents while meeting campus safety standards.

PROJECTS

SerenityStrings | *TypeScript, React, Python, FastAPI, Tailwind CSS, Tone.js, Axios*

- Developed a full-stack music theory and guitar-learning platform with an interactive fretboard, chord progression generator, scale explorer, metronome, and tuner.
- Built a responsive **React/TypeScript** fretboard with dynamic note mapping, scale visualization, chord highlighting, alternate tunings, and mobile-friendly interfaces.
- Implemented **Python/FastAPI** music-theory services for chord generation, scale construction, Roman numeral analysis, progression recommendations, audio playback, metronome timing, and pitch detection.

LuminosityLEDs | *TypeScript, Next.js, React, Node.js, Express, MongoDB, Docker, ActiveMQ*

- Collaborated on a **web application** for controlling **embedded** LED lights using **Next.js** and **React components**.
- Developed and maintained **Dockerized** microservices for an **ActiveMQ**-based notification system supporting message queuing and real-time notifications in a production cloud environment.
- Improved **MongoDB** connection handling across multiple services and collaborated using **Agile** practices, including **Kanban**, **Scrum**, and weekly team meetings.

SerenityNest | *Flutter, Dart, Firebase*

- Developed an **Android** well-being app featuring meditation, journaling, resources, and a quote-of-the-day page.
- Implemented email/password and **Google sign-in** with **Firebase Authentication** and **data storage**.

SKILLS, HACKATHONS & ORGANIZATIONS

Languages: TypeScript, JavaScript, HTML/CSS, Java, C++, Python, Dart
Technologies/Frameworks: React, Next.js, Node.js, Express, FastAPI, Tailwind CSS, MongoDB, MySQL, Firebase, Docker, Tone.js
Tools: Git/GitHub, Jira, Confluence, Figma, Notion, Vercel
Hackathons: CPP 2024 BroncoHacks, CPP 2025 BroncoHacks
Organizations: 2025 T. Howard Foundation Final Talent Pool Member